

\*57. Dynamic programming can be used to develop an algorithm for solving the matrix-chain multiplication problem introduced in Section 3.3. This is the problem of determining how the product  $A_1 A_2 \cdots A_n$  can be computed using the fewest integer multiplications, where  $A_1, A_2, \dots, A_n$  are  $m_1 \times m_2, m_2 \times m_3, \dots, m_n \times m_{n+1}$  matrices, respectively, and each matrix has integer entries. Recall that by the associative law, the product does not depend on the order in which the matrices are multiplied.

- a) Show that the brute-force method of determining the minimum number of integer multiplications needed to solve a matrix-chain multiplication problem has exponential worst-case complexity. [Hint: Do this by first showing that the order of multiplication of matrices is specified by parenthesizing the product. Then, use Example 5 and the result of part (c) of Exercise 43 in Section 8.4.]
- b) Denote by  $A_{ij}$  the product  $A_i A_{i+1} \cdots A_j$ , and  $M(i, j)$  the minimum number of integer multiplications required to find  $A_{ij}$ . Show that if the

least number of integer multiplications are used to compute  $A_{ij}$ , where  $i < j$ , by splitting the product into the product of  $A_i$  through  $A_k$  and the product of  $A_{k+1}$  through  $A_j$ , then the first  $k$  terms must be parenthesized so that  $A_{ik}$  is computed in the optimal way using  $M(i, k)$  integer multiplications, and  $A_{k+1,j}$  must be parenthesized so that  $A_{k+1,j}$  is computed in the optimal way using  $M(k+1, j)$  integer multiplications.

- c) Explain why part (b) leads to the recurrence relation  $M(i, j) = \min_{i \leq k < j} (M(i, k) + M(k+1, j) + m_i m_{k+1} m_{j+1})$  if  $1 \leq i \leq j < n$ .
- d) Use the recurrence relation in part (c) to construct an efficient algorithm for determining the order the  $n$  matrices should be multiplied to use the minimum number of integer multiplications. Store the partial results  $M(i, j)$  as you find them so that your algorithm will not have exponential complexity.
- e) Show that your algorithm from part (d) has  $O(n^3)$  worst-case complexity in terms of multiplications of integers.

## 8.2 Solving Linear Recurrence Relations

### 8.2.1 Introduction



A wide variety of recurrence relations occur in models. Some of these recurrence relations can be solved using iteration or some other ad hoc technique. However, one important class of recurrence relations can be explicitly solved in a systematic way. These are recurrence relations that express the terms of a sequence as linear combinations of previous terms.

#### Definition 1

A linear homogeneous recurrence relation of degree  $k$  with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k},$$

where  $c_1, c_2, \dots, c_k$  are real numbers, and  $c_k \neq 0$ .

The recurrence relation in the definition is **linear** because the right-hand side is a sum of previous terms of the sequence each multiplied by a function of  $n$ . The recurrence relation is **homogeneous** because no terms occur that are not multiples of the  $a_j$ s. The coefficients of the terms of the sequence are all **constants**, rather than functions that depend on  $n$ . The **degree** is  $k$  because  $a_n$  is expressed in terms of the previous  $k$  terms of the sequence.

A consequence of the second principle of mathematical induction is that a sequence satisfying the recurrence relation in the definition is uniquely determined by this recurrence relation and the  $k$  initial conditions

$$a_0 = C_0, a_1 = C_1, \dots, a_{k-1} = C_{k-1}.$$

#### EXAMPLE 1

The recurrence relation  $P_n = (1.11)P_{n-1}$  is a linear homogeneous recurrence relation of degree one. The recurrence relation  $f_n = f_{n-1} + f_{n-2}$  is a linear homogeneous recurrence relation of

degree two. The recurrence relation  $a_n = a_{n-5}$  is a linear homogeneous recurrence relation of degree five. ◀

To help clarify the definition of linear homogeneous recurrence relations with constant coefficients, we will now provide examples of recurrence relations each lacking one of the defining properties.

**EXAMPLE 2** The recurrence relation  $a_n = a_{n-1} + a_{n-2}^2$  is not linear. The recurrence relation  $H_n = 2H_{n-1} + 1$  is not homogeneous. The recurrence relation  $B_n = nB_{n-1}$  does not have constant coefficients. ◀

Linear homogeneous recurrence relations are studied for two reasons. First, they often occur in modeling of problems. Second, they can be systematically solved.

## 8.2.2 Solving Linear Homogeneous Recurrence Relations with Constant Coefficients

Recurrence relations may be difficult to solve, but fortunately this is not the case for linear homogeneous recurrence relations with constant coefficients. We can use two key ideas to find all their solutions. First, these recurrence relations have solutions of the form  $a_n = r^n$ , where  $r$  is a constant. To see this, observe that  $a_n = r^n$  is a solution of the recurrence relation  $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$  if and only if

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k}.$$

When both sides of this equation are divided by  $r^{n-k}$  (when  $r \neq 0$ ) and the right-hand side is subtracted from the left, we obtain the equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0.$$

Consequently, the sequence  $\{a_n\}$  with  $a_n = r^n$  where  $r \neq 0$  is a solution if and only if  $r$  is a solution of this last equation. We call this the **characteristic equation** of the recurrence relation. The solutions of this equation are called the **characteristic roots** of the recurrence relation. As we will see, these characteristic roots can be used to give an explicit formula for all the solutions of the recurrence relation.

The other key observation is that a linear combination of two solutions of a linear homogeneous recurrence relation is also a solution. To see this, suppose that  $s_n$  and  $t_n$  are both solutions of  $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ . Then

$$s_n = c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k}$$

and

$$t_n = c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}.$$

Now suppose that  $b_1$  and  $b_2$  are real numbers. Then

$$\begin{aligned} b_1 s_n + b_2 t_n &= b_1 (c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k}) + b_2 (c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}) \\ &= c_1 (b_1 s_{n-1} + b_2 t_{n-1}) + c_2 (b_1 s_{n-2} + b_2 t_{n-2}) + \cdots + c_k (b_1 s_{n-k} + b_2 t_{n-k}). \end{aligned}$$

This means that  $b_1 s_n + b_2 t_n$  is also a solution of the same linear homogeneous recurrence relation.

Using these key observations, we will show how to solve linear homogeneous recurrence relations with constant coefficients.

**THE DEGREE TWO CASE** We now turn our attention to linear homogeneous recurrence relations of degree two. First, consider the case when there are two distinct characteristic roots.

### THEOREM 1

Let  $c_1$  and  $c_2$  be real numbers. Suppose that  $r^2 - c_1r - c_2 = 0$  has two distinct roots  $r_1$  and  $r_2$ . Then the sequence  $\{a_n\}$  is a solution of the recurrence relation  $a_n = c_1a_{n-1} + c_2a_{n-2}$  if and only if  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  for  $n = 0, 1, 2, \dots$ , where  $\alpha_1$  and  $\alpha_2$  are constants.

**Proof:** We must do two things to prove the theorem. First, it must be shown that if  $r_1$  and  $r_2$  are the roots of the characteristic equation, and  $\alpha_1$  and  $\alpha_2$  are constants, then the sequence  $\{a_n\}$  with  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  is a solution of the recurrence relation. Second, it must be shown that if the sequence  $\{a_n\}$  is a solution, then  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  for some constants  $\alpha_1$  and  $\alpha_2$ .

We now show that if  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$ , then the sequence  $\{a_n\}$  is a solution of the recurrence relation. Because  $r_1$  and  $r_2$  are roots of  $r^2 - c_1r - c_2 = 0$ , it follows that  $r_1^2 = c_1r_1 + c_2$  and  $r_2^2 = c_1r_2 + c_2$ .

From these equations, we see that

$$\begin{aligned} c_1a_{n-1} + c_2a_{n-2} &= c_1(\alpha_1r_1^{n-1} + \alpha_2r_2^{n-1}) + c_2(\alpha_1r_1^{n-2} + \alpha_2r_2^{n-2}) \\ &= \alpha_1r_1^{n-2}(c_1r_1 + c_2) + \alpha_2r_2^{n-2}(c_1r_2 + c_2) \\ &= \alpha_1r_1^{n-2}r_1^2 + \alpha_2r_2^{n-2}r_2^2 \\ &= \alpha_1r_1^n + \alpha_2r_2^n \\ &= a_n. \end{aligned}$$

This shows that the sequence  $\{a_n\}$  with  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  is a solution of the recurrence relation.

To show that every solution  $\{a_n\}$  of the recurrence relation  $a_n = c_1a_{n-1} + c_2a_{n-2}$  has  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  for  $n = 0, 1, 2, \dots$ , for some constants  $\alpha_1$  and  $\alpha_2$ , suppose that  $\{a_n\}$  is a solution of the recurrence relation, and the initial conditions  $a_0 = C_0$  and  $a_1 = C_1$  hold. It will be shown that there are constants  $\alpha_1$  and  $\alpha_2$  such that the sequence  $\{a_n\}$  with  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  satisfies these same initial conditions. This requires that

$$\begin{aligned} a_0 &= C_0 = \alpha_1 + \alpha_2, \\ a_1 &= C_1 = \alpha_1r_1 + \alpha_2r_2. \end{aligned}$$

We can solve these two equations for  $\alpha_1$  and  $\alpha_2$ . From the first equation it follows that  $\alpha_2 = C_0 - \alpha_1$ . Inserting this expression into the second equation gives

$$C_1 = \alpha_1r_1 + (C_0 - \alpha_1)r_2.$$

Hence,

$$C_1 = \alpha_1(r_1 - r_2) + C_0r_2.$$

This shows that

$$\alpha_1 = \frac{C_1 - C_0r_2}{r_1 - r_2}$$

and

$$\alpha_2 = C_0 - \alpha_1 = C_0 - \frac{C_1 - C_0r_2}{r_1 - r_2} = \frac{C_0r_1 - C_1}{r_1 - r_2},$$

where these expressions for  $\alpha_1$  and  $\alpha_2$  depend on the fact that  $r_1 \neq r_2$ . (When  $r_1 = r_2$ , this theorem is not true.) Hence, with these values for  $\alpha_1$  and  $\alpha_2$ , the sequence  $\{a_n\}$  with  $\alpha_1 r_1^n + \alpha_2 r_2^n$  satisfies the two initial conditions.

We know that  $\{a_n\}$  and  $\{\alpha_1 r_1^n + \alpha_2 r_2^n\}$  are both solutions of the recurrence relation  $a_n = c_1 a_{n-1} + c_2 a_{n-2}$  and both satisfy the initial conditions when  $n = 0$  and  $n = 1$ . Because there is a unique solution of a linear homogeneous recurrence relation of degree two with two initial conditions, it follows that the two solutions are the same, that is,  $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$  for all nonnegative integers  $n$ . We have completed the proof by showing that a solution of the linear homogeneous recurrence relation with constant coefficients of degree two must be of the form  $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ , where  $\alpha_1$  and  $\alpha_2$  are constants. 

The characteristic roots of a linear homogeneous recurrence relation with constant coefficients may be complex numbers. Theorem 1 (and also subsequent theorems in this section) still applies in this case. Recurrence relations with complex characteristic roots will not be discussed in the text. Readers familiar with complex numbers may wish to solve Exercises 38 and 39.

Examples 3 and 4 show how to use Theorem 1 to solve recurrence relations.

### EXAMPLE 3 What is the solution of the recurrence relation

**Extra Examples** 

$$a_n = a_{n-1} + 2a_{n-2}$$

with  $a_0 = 2$  and  $a_1 = 7$ ?

**Solution:** Theorem 1 can be used to solve this problem. The characteristic equation of the recurrence relation is  $r^2 - r - 2 = 0$ . Its roots are  $r = 2$  and  $r = -1$ . Hence, the sequence  $\{a_n\}$  is a solution to the recurrence relation if and only if

$$a_n = \alpha_1 2^n + \alpha_2 (-1)^n,$$

for some constants  $\alpha_1$  and  $\alpha_2$ . From the initial conditions, it follows that

$$\begin{aligned} a_0 = 2 &= \alpha_1 + \alpha_2, \\ a_1 = 7 &= \alpha_1 \cdot 2 + \alpha_2 \cdot (-1). \end{aligned}$$

Solving these two equations shows that  $\alpha_1 = 3$  and  $\alpha_2 = -1$ . Hence, the solution to the recurrence relation and initial conditions is the sequence  $\{a_n\}$  with

$$a_n = 3 \cdot 2^n - (-1)^n. \quad \text{◀}$$

### EXAMPLE 4 Find an explicit formula for the Fibonacci numbers.

**Solution:** Recall that the sequence of Fibonacci numbers satisfies the recurrence relation  $f_n = f_{n-1} + f_{n-2}$  and also satisfies the initial conditions  $f_0 = 0$  and  $f_1 = 1$ . The roots of the characteristic equation  $r^2 - r - 1 = 0$  are  $r_1 = (1 + \sqrt{5})/2$  and  $r_2 = (1 - \sqrt{5})/2$ . Therefore, from Theorem 1 it follows that the Fibonacci numbers are given by

$$f_n = \alpha_1 \left( \frac{1 + \sqrt{5}}{2} \right)^n + \alpha_2 \left( \frac{1 - \sqrt{5}}{2} \right)^n,$$

for some constants  $\alpha_1$  and  $\alpha_2$ . The initial conditions  $f_0 = 0$  and  $f_1 = 1$  can be used to find these constants. We have

$$\begin{aligned} f_0 &= \alpha_1 + \alpha_2 = 0, \\ f_1 &= \alpha_1 \left( \frac{1 + \sqrt{5}}{2} \right) + \alpha_2 \left( \frac{1 - \sqrt{5}}{2} \right) = 1. \end{aligned}$$

The solution to these simultaneous equations for  $\alpha_1$  and  $\alpha_2$  is

$$\alpha_1 = 1/\sqrt{5}, \quad \alpha_2 = -1/\sqrt{5}.$$

Consequently, the Fibonacci numbers are given by

$$f_n = \frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left( \frac{1 - \sqrt{5}}{2} \right)^n.$$

Theorem 1 does not apply when there is one characteristic root of multiplicity two. If this happens, then  $a_n = nr_0^n$  is another solution of the recurrence relation when  $r_0$  is a root of multiplicity two of the characteristic equation. Theorem 2 shows how to handle this case.

### THEOREM 2

Let  $c_1$  and  $c_2$  be real numbers with  $c_2 \neq 0$ . Suppose that  $r^2 - c_1r - c_2 = 0$  has only one root  $r_0$ . A sequence  $\{a_n\}$  is a solution of the recurrence relation  $a_n = c_1a_{n-1} + c_2a_{n-2}$  if and only if  $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$ , for  $n = 0, 1, 2, \dots$ , where  $\alpha_1$  and  $\alpha_2$  are constants.

The proof of Theorem 2 is left as Exercise 10. Example 5 illustrates the use of this theorem.

### EXAMPLE 5

What is the solution of the recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2}$$

with initial conditions  $a_0 = 1$  and  $a_1 = 6$ ?

**Solution:** The only root of  $r^2 - 6r + 9 = 0$  is  $r = 3$ . Hence, the solution to this recurrence relation is

$$a_n = \alpha_1 3^n + \alpha_2 n 3^n$$

for some constants  $\alpha_1$  and  $\alpha_2$ . Using the initial conditions, it follows that

$$\begin{aligned} a_0 &= 1 = \alpha_1, \\ a_1 &= 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 3. \end{aligned}$$

Solving these two equations shows that  $\alpha_1 = 1$  and  $\alpha_2 = 1$ . Consequently, the solution to this recurrence relation and the initial conditions is

$$a_n = 3^n + n3^n.$$

**THE GENERAL CASE** We will now state the general result about the solution of linear homogeneous recurrence relations with constant coefficients, where the degree may be greater than two, under the assumption that the characteristic equation has distinct roots. The proof of this result will be left as Exercise 16.

**THEOREM 3**

Let  $c_1, c_2, \dots, c_k$  be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has  $k$  distinct roots  $r_1, r_2, \dots, r_k$ . Then a sequence  $\{a_n\}$  is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

for  $n = 0, 1, 2, \dots$ , where  $\alpha_1, \alpha_2, \dots, \alpha_k$  are constants.

We illustrate the use of the theorem with Example 6.

**EXAMPLE 6** Find the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions  $a_0 = 2$ ,  $a_1 = 5$ , and  $a_2 = 15$ .

**Solution:** The characteristic polynomial of this recurrence relation is

$$r^3 - 6r^2 + 11r - 6.$$

The characteristic roots are  $r = 1$ ,  $r = 2$ , and  $r = 3$ , because  $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3)$ . Hence, the solutions to this recurrence relation are of the form

$$a_n = \alpha_1 \cdot 1^n + \alpha_2 \cdot 2^n + \alpha_3 \cdot 3^n.$$

To find the constants  $\alpha_1$ ,  $\alpha_2$ , and  $\alpha_3$ , use the initial conditions. This gives

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3,$$

$$a_1 = 5 = \alpha_1 + \alpha_2 \cdot 2 + \alpha_3 \cdot 3,$$

$$a_2 = 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9.$$

When these three simultaneous equations are solved for  $\alpha_1$ ,  $\alpha_2$ , and  $\alpha_3$ , we find that  $\alpha_1 = 1$ ,  $\alpha_2 = -1$ , and  $\alpha_3 = 2$ . Hence, the unique solution to this recurrence relation and the given initial conditions is the sequence  $\{a_n\}$  with

$$a_n = 1 - 2^n + 2 \cdot 3^n.$$

We now state the most general result about linear homogeneous recurrence relations with constant coefficients, allowing the characteristic equation to have multiple roots. The key point is that for each root  $r$  of the characteristic equation, the general solution has a summand of the

form  $P(n)r^n$ , where  $P(n)$  is a polynomial of degree  $m - 1$ , with  $m$  the multiplicity of this root. We leave the proof of this result as Exercise 51.

**THEOREM 4**

Let  $c_1, c_2, \dots, c_k$  be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has  $t$  distinct roots  $r_1, r_2, \dots, r_t$  with multiplicities  $m_1, m_2, \dots, m_t$ , respectively, so that  $m_i \geq 1$  for  $i = 1, 2, \dots, t$  and  $m_1 + m_2 + \dots + m_t = k$ . Then a sequence  $\{a_n\}$  is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for  $n = 0, 1, 2, \dots$ , where  $\alpha_{i,j}$  are constants for  $1 \leq i \leq t$  and  $0 \leq j \leq m_i - 1$ .

Example 7 illustrates how Theorem 4 is used to find the general form of a solution of a linear homogeneous recurrence relation when the characteristic equation has several repeated roots.

**EXAMPLE 7**

Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one). What is the form of the general solution?

**Solution:** By Theorem 4, the general form of the solution is

$$(\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2)2^n + (\alpha_{2,0} + \alpha_{2,1}n)5^n + \alpha_{3,0}9^n.$$

We now illustrate the use of Theorem 4 to solve a linear homogeneous recurrence relation with constant coefficients when the characteristic equation has a root of multiplicity three.

**EXAMPLE 8**

Find the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

with initial conditions  $a_0 = 1$ ,  $a_1 = -2$ , and  $a_2 = -1$ .

**Solution:** The characteristic equation of this recurrence relation is

$$r^3 + 3r^2 + 3r + 1 = 0.$$

Because  $r^3 + 3r^2 + 3r + 1 = (r + 1)^3$ , there is a single root  $r = -1$  of multiplicity three of the characteristic equation. By Theorem 4 the solutions of this recurrence relation are of the form

$$a_n = \alpha_{1,0}(-1)^n + \alpha_{1,1}n(-1)^n + \alpha_{1,2}n^2(-1)^n.$$

To find the constants  $\alpha_{1,0}$ ,  $\alpha_{1,1}$ , and  $\alpha_{1,2}$ , use the initial conditions. This gives

$$\begin{aligned} a_0 &= 1 = \alpha_{1,0}, \\ a_1 &= -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,2}, \\ a_2 &= -1 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,2}. \end{aligned}$$

The simultaneous solution of these three equations is  $\alpha_{1,0} = 1$ ,  $\alpha_{1,1} = 3$ , and  $\alpha_{1,2} = -2$ . Hence, the unique solution to this recurrence relation and the given initial conditions is the sequence  $\{a_n\}$  with

$$a_n = (1 + 3n - 2n^2)(-1)^n.$$

### 8.2.3 Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

We have seen how to solve linear homogeneous recurrence relations with constant coefficients. Is there a relatively simple technique for solving a linear, but not homogeneous, recurrence relation with constant coefficients, such as  $a_n = 3a_{n-1} + 2n$ ? We will see that the answer is yes for certain families of such recurrence relations.

The recurrence relation  $a_n = 3a_{n-1} + 2n$  is an example of a **linear nonhomogeneous recurrence relation with constant coefficients**, that is, a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n),$$

where  $c_1, c_2, \dots, c_k$  are real numbers and  $F(n)$  is a function not identically zero depending only on  $n$ . The recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

is called the **associated homogeneous recurrence relation**. It plays an important role in the solution of the nonhomogeneous recurrence relation.

**EXAMPLE 9** Each of the recurrence relations  $a_n = a_{n-1} + 2^n$ ,  $a_n = a_{n-1} + a_{n-2} + n^2 + n + 1$ ,  $a_n = 3a_{n-1} + n3^n$ , and  $a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$  is a linear nonhomogeneous recurrence relation with constant coefficients. The associated linear homogeneous recurrence relations are  $a_n = a_{n-1}$ ,  $a_n = a_{n-1} + a_{n-2}$ ,  $a_n = 3a_{n-1}$ , and  $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ , respectively.

The key fact about linear nonhomogeneous recurrence relations with constant coefficients is that every solution is the sum of a particular solution and a solution of the associated linear homogeneous recurrence relation, as Theorem 5 shows.

#### THEOREM 5

If  $\{a_n^{(p)}\}$  is a particular solution of the nonhomogeneous linear recurrence relation with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n),$$

then every solution is of the form  $\{a_n^{(p)} + a_n^{(h)}\}$ , where  $\{a_n^{(h)}\}$  is a solution of the associated homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}.$$



**Proof:** Because  $\{a_n^{(p)}\}$  is a particular solution of the nonhomogeneous recurrence relation, we know that

$$a_n^{(p)} = c_1 a_{n-1}^{(p)} + c_2 a_{n-2}^{(p)} + \cdots + c_k a_{n-k}^{(p)} + F(n).$$

Now suppose that  $\{b_n\}$  is a second solution of the nonhomogeneous recurrence relation, so that

$$b_n = c_1 b_{n-1} + c_2 b_{n-2} + \cdots + c_k b_{n-k} + F(n).$$

Subtracting the first of these two equations from the second shows that

$$b_n - a_n^{(p)} = c_1 (b_{n-1} - a_{n-1}^{(p)}) + c_2 (b_{n-2} - a_{n-2}^{(p)}) + \cdots + c_k (b_{n-k} - a_{n-k}^{(p)}).$$

It follows that  $\{b_n - a_n^{(p)}\}$  is a solution of the associated homogeneous linear recurrence, say,  $\{a_n^{(h)}\}$ . Consequently,  $b_n = a_n^{(p)} + a_n^{(h)}$  for all  $n$ . 

By Theorem 5, we see that the key to solving nonhomogeneous recurrence relations with constant coefficients is finding a particular solution. Then every solution is a sum of this solution and a solution of the associated homogeneous recurrence relation. Although there is no general method for finding such a solution that works for every function  $F(n)$ , there are techniques that work for certain types of functions  $F(n)$ , such as polynomials and powers of constants. This is illustrated in Examples 10 and 11.

**EXAMPLE 10** Find all solutions of the recurrence relation  $a_n = 3a_{n-1} + 2n$ . What is the solution with  $a_1 = 3$ ?

**Solution:** To solve this linear nonhomogeneous recurrence relation with constant coefficients, we need to solve its associated linear homogeneous equation and to find a particular solution for the given nonhomogeneous equation. The associated linear homogeneous equation is  $a_n = 3a_{n-1}$ . Its solutions are  $a_n^{(h)} = \alpha 3^n$ , where  $\alpha$  is a constant.

We now find a particular solution. Because  $F(n) = 2n$  is a polynomial in  $n$  of degree one, a reasonable trial solution is a linear function in  $n$ , say,  $p_n = cn + d$ , where  $c$  and  $d$  are constants. To determine whether there are any solutions of this form, suppose that  $p_n = cn + d$  is such a solution. Then the equation  $a_n = 3a_{n-1} + 2n$  becomes  $cn + d = 3(c(n-1) + d) + 2n$ . Simplifying and combining like terms gives  $(2 + 2c)n + (2d - 3c) = 0$ . It follows that  $cn + d$  is a solution if and only if  $2 + 2c = 0$  and  $2d - 3c = 0$ . This shows that  $cn + d$  is a solution if and only if  $c = -1$  and  $d = -3/2$ . Consequently,  $a_n^{(p)} = -n - 3/2$  is a particular solution.

By Theorem 5 all solutions are of the form

$$a_n = a_n^{(p)} + a_n^{(h)} = -n - \frac{3}{2} + \alpha \cdot 3^n,$$

where  $\alpha$  is a constant.

To find the solution with  $a_1 = 3$ , let  $n = 1$  in the formula we obtained for the general solution. We find that  $3 = -1 - 3/2 + 3\alpha$ , which implies that  $\alpha = 11/6$ . The solution we seek is  $a_n = -n - 3/2 + (11/6)3^n$ . 

**EXAMPLE 11** Find all solutions of the recurrence relation

**Extra  
Examples** 

$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n.$$

**Solution:** This is a linear nonhomogeneous recurrence relation. The solutions of its associated homogeneous recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2}$$

are  $a_n^{(h)} = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n$ , where  $\alpha_1$  and  $\alpha_2$  are constants. Because  $F(n) = 7^n$ , a reasonable trial solution is  $a_n^{(p)} = C \cdot 7^n$ , where  $C$  is a constant. Substituting the terms of this sequence into the recurrence relation implies that  $C \cdot 7^n = 5C \cdot 7^{n-1} - 6C \cdot 7^{n-2} + 7^n$ . Factoring out  $7^{n-2}$ , this equation becomes  $49C = 35C - 6C + 49$ , which implies that  $20C = 49$ , or that  $C = 49/20$ . Hence,  $a_n^{(p)} = (49/20)7^n$  is a particular solution. By Theorem 5, all solutions are of the form

$$a_n = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n + (49/20)7^n.$$

In Examples 10 and 11, we made an educated guess that there are solutions of a particular form. In both cases we were able to find particular solutions. This was not an accident. Whenever  $F(n)$  is the product of a polynomial in  $n$  and the  $n$ th power of a constant, we know exactly what form a particular solution has, as stated in Theorem 6. We leave the proof of Theorem 6 as Exercise 52.

### THEOREM 6

Suppose that  $\{a_n\}$  satisfies the linear nonhomogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n),$$

where  $c_1, c_2, \dots, c_k$  are real numbers, and

$$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \cdots + b_1 n + b_0) s^n,$$

where  $b_0, b_1, \dots, b_t$  and  $s$  are real numbers. When  $s$  is not a root of the characteristic equation of the associated linear homogeneous recurrence relation, there is a particular solution of the form

$$(p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0) s^n.$$

When  $s$  is a root of this characteristic equation and its multiplicity is  $m$ , there is a particular solution of the form

$$n^m (p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0) s^n.$$

Note that in the case when  $s$  is a root of multiplicity  $m$  of the characteristic equation of the associated linear homogeneous recurrence relation, the factor  $n^m$  ensures that the proposed particular solution will not already be a solution of the associated linear homogeneous recurrence relation. We next provide Example 12 to illustrate the form of a particular solution provided by Theorem 6.

### EXAMPLE 12

What form does a particular solution of the linear nonhomogeneous recurrence relation  $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$  have when  $F(n) = 3^n$ ,  $F(n) = n3^n$ ,  $F(n) = n^2 2^n$ , and  $F(n) = (n^2 + 1)3^n$ ?

**Solution:** The associated linear homogeneous recurrence relation is  $a_n = 6a_{n-1} - 9a_{n-2}$ . Its characteristic equation,  $r^2 - 6r + 9 = (r - 3)^2 = 0$ , has a single root, 3, of multiplicity two. To apply Theorem 6, with  $F(n)$  of the form  $P(n)s^n$ , where  $P(n)$  is a polynomial and  $s$  is a constant, we need to ask whether  $s$  is a root of this characteristic equation.

Because  $s = 3$  is a root with multiplicity  $m = 2$  but  $s = 2$  is not a root, Theorem 6 tells us that a particular solution has the form  $p_0 n^2 3^n$  if  $F(n) = 3^n$ , the form  $n^2(p_1 n + p_0)3^n$  if  $F(n) =$

$n3^n$ , the form  $(p_2n^2 + p_1n + p_0)2^n$  if  $F(n) = n^22^n$ , and the form  $n^2(p_2n^2 + p_1n + p_0)3^n$  if  $F(n) = (n^2 + 1)3^n$ . 

Care must be taken when  $s = 1$  when solving recurrence relations of the type covered by Theorem 6. In particular, to apply this theorem with  $F(n) = b_in_t + b_{t-1}n_{t-1} + \cdots + b_1n + b_0$ , the parameter  $s$  takes the value  $s = 1$  (even though the term  $1^n$  does not explicitly appear). By the theorem, the form of the solution then depends on whether 1 is a root of the characteristic equation of the associated linear homogeneous recurrence relation. This is illustrated in Example 13, which shows how Theorem 6 can be used to find a formula for the sum of the first  $n$  positive integers.

**EXAMPLE 13** Let  $a_n$  be the sum of the first  $n$  positive integers, so that

$$a_n = \sum_{k=1}^n k.$$

Note that  $a_n$  satisfies the linear nonhomogeneous recurrence relation

$$a_n = a_{n-1} + n.$$

(To obtain  $a_n$ , the sum of the first  $n$  positive integers, from  $a_{n-1}$ , the sum of the first  $n - 1$  positive integers, we add  $n$ .) Note that the initial condition is  $a_1 = 1$ .

The associated linear homogeneous recurrence relation for  $a_n$  is

$$a_n = a_{n-1}.$$

The solutions of this homogeneous recurrence relation are given by  $a_n^{(h)} = c(1)^n = c$ , where  $c$  is a constant. To find all solutions of  $a_n = a_{n-1} + n$ , we need find only a single particular solution. By Theorem 6, because  $F(n) = n = n \cdot (1)^n$  and  $s = 1$  is a root of degree one of the characteristic equation of the associated linear homogeneous recurrence relation, there is a particular solution of the form  $n(p_1n + p_0) = p_1n^2 + p_0n$ .

Inserting this into the recurrence relation gives  $p_1n^2 + p_0n = p_1(n-1)^2 + p_0(n-1) + n$ . Simplifying, we see that  $n(2p_1 - 1) + (p_0 - p_1) = 0$ , which means that  $2p_1 - 1 = 0$  and  $p_0 - p_1 = 0$ , so  $p_0 = p_1 = 1/2$ . Hence,

$$a_n^{(p)} = \frac{n^2}{2} + \frac{n}{2} = \frac{n(n+1)}{2}$$

is a particular solution. Hence, all solutions of the original recurrence relation  $a_n = a_{n-1} + n$  are given by  $a_n = a_n^{(h)} + a_n^{(p)} = c + n(n+1)/2$ . Because  $a_1 = 1$ , we have  $1 = a_1 = c + 1 \cdot 2/2 = c + 1$ , so  $c = 0$ . It follows that  $a_n = n(n+1)/2$ . (This is the same formula given in Table 2 in Section 2.4 and derived previously.) 

## Exercises

- Determine which of these are linear homogeneous recurrence relations with constant coefficients. Also, find the degree of those that are.
  - $a_n = 3a_{n-1} + 4a_{n-2} + 5a_{n-3}$
  - $a_n = 2na_{n-1} + a_{n-2}$
  - $a_n = a_{n-1} + a_{n-4}$
  - $a_n = a_{n-1} + 2$
  - $a_n = a_{n-1}^2 + a_{n-2}$
  - $a_n = a_{n-1} + n$
- Determine which of these are linear homogeneous recurrence relations with constant coefficients. Also, find the degree of those that are.
  - $a_n = 3a_{n-2}$
  - $a_n = 3$
  - $a_n = a_{n-1}^2/n$
  - $a_n = a_{n-1} + 2a_{n-3}$
  - $a_n = a_{n-1} + a_{n-2} + n + 3$
  - $a_n = 4a_{n-2} + 5a_{n-4} + 9a_{n-7}$

3. Solve these recurrence relations together with the initial conditions given.

- a)  $a_n = 2a_{n-1}$  for  $n \geq 1$ ,  $a_0 = 3$
- b)  $a_n = a_{n-1}$  for  $n \geq 1$ ,  $a_0 = 2$
- c)  $a_n = 5a_{n-1} - 6a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 1$ ,  $a_1 = 0$
- d)  $a_n = 4a_{n-1} - 4a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 6$ ,  $a_1 = 8$
- e)  $a_n = -4a_{n-1} - 4a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 0$ ,  $a_1 = 1$
- f)  $a_n = 4a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 0$ ,  $a_1 = 4$
- g)  $a_n = a_{n-2}/4$  for  $n \geq 2$ ,  $a_0 = 1$ ,  $a_1 = 0$

4. Solve these recurrence relations together with the initial conditions given.

- a)  $a_n = a_{n-1} + 6a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 3$ ,  $a_1 = 6$
- b)  $a_n = 7a_{n-1} - 10a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 2$ ,  $a_1 = 1$
- c)  $a_n = 6a_{n-1} - 8a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 4$ ,  $a_1 = 10$
- d)  $a_n = 2a_{n-1} - a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 4$ ,  $a_1 = 1$
- e)  $a_n = a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 5$ ,  $a_1 = -1$
- f)  $a_n = -6a_{n-1} - 9a_{n-2}$  for  $n \geq 2$ ,  $a_0 = 3$ ,  $a_1 = -3$
- g)  $a_{n+2} = -4a_{n+1} + 5a_n$  for  $n \geq 0$ ,  $a_0 = 2$ ,  $a_1 = 8$

5. How many different messages can be transmitted in  $n$  microseconds using the two signals described in Exercise 19 in Section 8.1?

6. How many different messages can be transmitted in  $n$  microseconds using three different signals if one signal requires 1 microsecond for transmittal, the other two signals require 2 microseconds each for transmittal, and a signal in a message is followed immediately by the next signal?

7. In how many ways can a  $2 \times n$  rectangular checkerboard be tiled using  $1 \times 2$  and  $2 \times 2$  pieces?

8. A model for the number of lobsters caught per year is based on the assumption that the number of lobsters caught in a year is the average of the number caught in the two previous years.

- a) Find a recurrence relation for  $\{L_n\}$ , where  $L_n$  is the number of lobsters caught in year  $n$ , under the assumption for this model.
- b) Find  $L_n$  if 100,000 lobsters were caught in year 1 and 300,000 were caught in year 2.

9. A deposit of \$100,000 is made to an investment fund at the beginning of a year. On the last day of each year two dividends are awarded. The first dividend is 20% of the amount in the account during that year. The second dividend is 45% of the amount in the account in the previous year.

- a) Find a recurrence relation for  $\{P_n\}$ , where  $P_n$  is the amount in the account at the end of  $n$  years if no money is ever withdrawn.
- b) How much is in the account after  $n$  years if no money has been withdrawn?

- \*10. Prove Theorem 2.

11. The **Lucas numbers** satisfy the recurrence relation

**Links**  $L_n = L_{n-1} + L_{n-2},$

and the initial conditions  $L_0 = 2$  and  $L_1 = 1$ .

- a) Show that  $L_n = f_{n-1} + f_{n+1}$  for  $n = 2, 3, \dots$ , where  $f_n$  is the  $n$ th Fibonacci number.
- b) Find an explicit formula for the Lucas numbers.

12. Find the solution to  $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$  for  $n = 3, 4, 5, \dots$ , with  $a_0 = 3$ ,  $a_1 = 6$ , and  $a_2 = 0$ .

13. Find the solution to  $a_n = 7a_{n-2} + 6a_{n-3}$  with  $a_0 = 9$ ,  $a_1 = 10$ , and  $a_2 = 32$ .

14. Find the solution to  $a_n = 5a_{n-2} - 4a_{n-4}$  with  $a_0 = 3$ ,  $a_1 = 2$ ,  $a_2 = 6$ , and  $a_3 = 8$ .

15. Find the solution to  $a_n = 2a_{n-1} + 5a_{n-2} - 6a_{n-3}$  with  $a_0 = 7$ ,  $a_1 = -4$ , and  $a_2 = 8$ .

- \*16. Prove Theorem 3.

17. Prove this identity relating the Fibonacci numbers and the binomial coefficients:

$$f_{n+1} = C(n, 0) + C(n-1, 1) + \dots + C(n-k, k),$$

where  $n$  is a positive integer and  $k = \lfloor n/2 \rfloor$ . [Hint: Let  $a_n = C(n, 0) + C(n-1, 1) + \dots + C(n-k, k)$ . Show that the sequence  $\{a_n\}$  satisfies the same recurrence relation and initial conditions satisfied by the sequence of Fibonacci numbers.]

18. Solve the recurrence relation  $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$  with  $a_0 = -5$ ,  $a_1 = 4$ , and  $a_2 = 88$ .

19. Solve the recurrence relation  $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$  with  $a_0 = 5$ ,  $a_1 = -9$ , and  $a_2 = 15$ .

20. Find the general form of the solutions of the recurrence relation  $a_n = 8a_{n-2} - 16a_{n-4}$ .

21. What is the general form of the solutions of a linear homogeneous recurrence relation if its characteristic equation has roots 1, 1, 1, 1, -2, -2, -2, 3, 3, -4?

22. What is the general form of the solutions of a linear homogeneous recurrence relation if its characteristic equation has the roots -1, -1, -1, 2, 2, 5, 5, 7?

23. Consider the nonhomogeneous linear recurrence relation  $a_n = 3a_{n-1} + 2^n$ .

- a) Show that  $a_n = -2^{n+1}$  is a solution of this recurrence relation.

- b) Use Theorem 5 to find all solutions of this recurrence relation.

- c) Find the solution with  $a_0 = 1$ .

24. Consider the nonhomogeneous linear recurrence relation  $a_n = 2a_{n-1} + 2^n$ .

- a) Show that  $a_n = n2^n$  is a solution of this recurrence relation.

- b) Use Theorem 5 to find all solutions of this recurrence relation.

- c) Find the solution with  $a_0 = 2$ .

25. a) Determine values of the constants  $A$  and  $B$  such that  $a_n = An + B$  is a solution of recurrence relation  $a_n = 2a_{n-1} + n + 5$ .

- b) Use Theorem 5 to find all solutions of this recurrence relation.

- c) Find the solution of this recurrence relation with  $a_0 = 4$ .

26. What is the general form of the particular solution guaranteed to exist by Theorem 6 of the linear nonhomogeneous recurrence relation  $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3} + F(n)$  if
- $F(n) = n^2?$
  - $F(n) = 2^n?$
  - $F(n) = n2^n?$
  - $F(n) = (-2)^n?$
  - $F(n) = n^22^n?$
  - $F(n) = n^3(-2)^n?$
  - $F(n) = 3?$
27. What is the general form of the particular solution guaranteed to exist by Theorem 6 of the linear nonhomogeneous recurrence relation  $a_n = 8a_{n-2} - 16a_{n-4} + F(n)$  if
- $F(n) = n^3?$
  - $F(n) = (-2)^n?$
  - $F(n) = n2^n?$
  - $F(n) = n^24^n?$
  - $F(n) = (n^2 - 2)(-2)^n?$
  - $F(n) = n^42^n?$
  - $F(n) = 2?$
28. a) Find all solutions of the recurrence relation  $a_n = 2a_{n-1} + 2n^2$ .  
b) Find the solution of the recurrence relation in part (a) with initial condition  $a_1 = 4$ .
29. a) Find all solutions of the recurrence relation  $a_n = 2a_{n-1} + 3^n$ .  
b) Find the solution of the recurrence relation in part (a) with initial condition  $a_1 = 5$ .
30. a) Find all solutions of the recurrence relation  $a_n = -5a_{n-1} - 6a_{n-2} + 42 \cdot 4^n$ .  
b) Find the solution of this recurrence relation with  $a_1 = 56$  and  $a_2 = 278$ .
31. Find all solutions of the recurrence relation  $a_n = 5a_{n-1} - 6a_{n-2} + 2^n + 3n$ . [Hint: Look for a particular solution of the form  $qn2^n + p_1n + p_2$ , where  $q$ ,  $p_1$ , and  $p_2$  are constants.]
32. Find the solution of the recurrence relation  $a_n = 2a_{n-1} + 3 \cdot 2^n$ .
33. Find all solutions of the recurrence relation  $a_n = 4a_{n-1} - 4a_{n-2} + (n+1)2^n$ .
34. Find all solutions of the recurrence relation  $a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3} + n4^n$  with  $a_0 = -2$ ,  $a_1 = 0$ , and  $a_2 = 5$ .
35. Find the solution of the recurrence relation  $a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3$  with  $a_0 = 1$  and  $a_1 = 4$ .
36. Let  $a_n$  be the sum of the first  $n$  perfect squares, that is,  $a_n = \sum_{k=1}^n k^2$ . Show that the sequence  $\{a_n\}$  satisfies the linear nonhomogeneous recurrence relation  $a_n = a_{n-1} + n^2$  and the initial condition  $a_1 = 1$ . Use Theorem 6 to determine a formula for  $a_n$  by solving this recurrence relation.
37. Let  $a_n$  be the sum of the first  $n$  triangular numbers, that is,  $a_n = \sum_{k=1}^n t_k$ , where  $t_k = k(k+1)/2$ . Show that  $\{a_n\}$  satisfies the linear nonhomogeneous recurrence relation  $a_n = a_{n-1} + n(n+1)/2$  and the initial condition  $a_1 = 1$ . Use Theorem 6 to determine a formula for  $a_n$  by solving this recurrence relation.
38. a) Find the characteristic roots of the linear homogeneous recurrence relation  $a_n = 2a_{n-1} - 2a_{n-2}$ . [Note: These are complex numbers.]
- b) Find the solution of the recurrence relation in part (a) with  $a_0 = 1$  and  $a_1 = 2$ .
- \*39. a) Find the characteristic roots of the linear homogeneous recurrence relation  $a_n = a_{n-4}$ . [Note: These include complex numbers.]  
b) Find the solution of the recurrence relation in part (a) with  $a_0 = 1$ ,  $a_1 = 0$ ,  $a_2 = -1$ , and  $a_3 = 1$ .
- \*40. Solve the simultaneous recurrence relations
- $$a_n = 3a_{n-1} + 2b_{n-1}$$
- $$b_n = a_{n-1} + 2b_{n-1}$$
- with  $a_0 = 1$  and  $b_0 = 2$ .
- \*41. a) Use the formula found in Example 4 for  $f_n$ , the  $n$ th Fibonacci number, to show that  $f_n$  is the integer closest to
- $$\frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^n$$
- b) Determine for which  $n$   $f_n$  is greater than
- $$\frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^n$$
- and for which  $n$   $f_n$  is less than
- $$\frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^n$$
42. Show that if  $a_n = a_{n-1} + a_{n-2}$ ,  $a_0 = s$  and  $a_1 = t$ , where  $s$  and  $t$  are constants, then  $a_n = sf_{n-1} + tf_n$  for all positive integers  $n$ .
43. Express the solution of the linear nonhomogeneous recurrence relation  $a_n = a_{n-1} + a_{n-2} + 1$  for  $n \geq 2$  where  $a_0 = 0$  and  $a_1 = 1$  in terms of the Fibonacci numbers. [Hint: Let  $b_n = a_n + 1$  and apply Exercise 42 to the sequence  $b_n$ .]
- \*44. (Linear algebra required) Let  $\mathbf{A}_n$  be the  $n \times n$  matrix with 2s on its main diagonal, 1s in all positions next to a diagonal element, and 0s everywhere else. Find a recurrence relation for  $d_n$ , the determinant of  $\mathbf{A}_n$ . Solve this recurrence relation to find a formula for  $d_n$ .
45. Suppose that each pair of a genetically engineered species of rabbits left on an island produces two new pairs of rabbits at the age of 1 month and six new pairs of rabbits at the age of 2 months and every month afterward. None of the rabbits ever die or leave the island.
- Find a recurrence relation for the number of pairs of rabbits on the island  $n$  months after one newborn pair is left on the island.
  - By solving the recurrence relation in (a) determine the number of pairs of rabbits on the island  $n$  months after one pair is left on the island.
46. Suppose that there are two goats on an island initially. The number of goats on the island doubles every year by natural reproduction, and some goats are either added or removed each year.